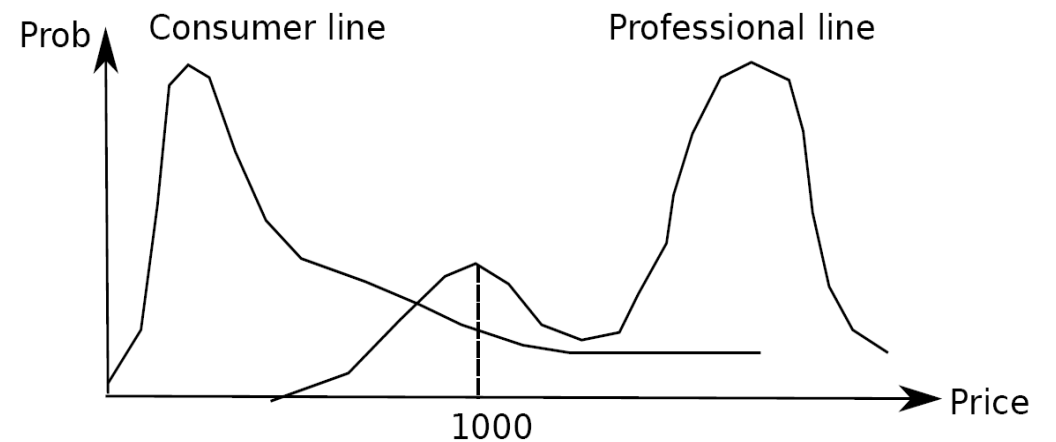


Probabilistic Model-Based Clustering

- ❑ Cluster analysis is to find hidden categories
- ❑ A hidden category (i.e., probabilistic cluster) is a distribution over the data space, which can be mathematically represented using a probability density function (or distribution function)
- ❑ Example: two categories for digital cameras sold
 - ❑ Consumer line vs. professional line
 - ❑ Density functions f_1, f_2 for C_1, C_2
 - ❑ Obtained by probabilistic clustering



Probabilistic Model-Based Clustering

- A mixture model assumes that a set of observed objects is a mixture of instances from multiple probabilistic clusters, and conceptually each observed object is generated independently
- Our task: infer a set of k probabilistic clusters that is mostly likely to generate D using the above data generation process
 - A set C of k probabilistic clusters $C_1, \dots, C_k$ with probability density functions $f_1, \dots, f_k$, respectively, and their probabilities $\omega_1, \dots, \omega_k$
 - Probability of an object o generated by cluster C_j is $P(o|C_j) = \omega_j f_j(o)$
 - Probability of o generated by the set of clusters C is $P(o|C) = \sum_{j=1}^k \omega_j f_j(o)$

Model-Based Clustering

- Since objects are assumed to be generated independently, for a data set $D = \{o_1, \dots, o_n\}$, we have,

$$P(D|C) = \prod_{i=1}^n P(o_i|C) = \prod_{i=1}^n \sum_{j=1}^k \omega_j f_j(o_i)$$

- Task: Find a set C of k probabilistic clusters such that $P(D|C)$ is maximized
- However, maximizing $P(D|C)$ is often intractable since the probability density function of a cluster can take an arbitrarily complicated form
- To make it computationally feasible (as a compromise), assume the probability density functions being some parameterized distributions

Univariate Gaussian Mixture Model

- $O = \{o_1, \dots, o_n\}$ (n observed objects), $\Theta = \{\theta_1, \dots, \theta_k\}$ (parameters of the k distributions), and $P_j(o_i | \theta_j)$ is the probability that o_i is generated from the j-th distribution using parameter θ_j , we have

$$P(o_i | \Theta) = \sum_{j=1}^k \omega_j P_j(o_i | \theta_j)$$

- Thus,

$$P(O | \Theta) = \prod_{i=1}^n \sum_{j=1}^k \omega_j P_j(o_i | \theta_j)$$

Univariate Gaussian Mixture Model

- Assume the probability density function of each cluster follows a 1-d Gaussian distribution. Suppose that there are k clusters
- The probability density function of each cluster are centered at μ_j with standard deviation σ_j , $\theta_j = (\mu_j, \sigma_j)$, we have

$$P(o_i | \theta_j) = \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(o_i - \mu_j)^2}{2\sigma_j^2}}$$

$$P(o_i | \Theta) = \sum_{j=1}^k \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(o_i - \mu_j)^2}{2\sigma_j^2}}$$

$$P(O | \Theta) = \prod_{i=1}^n \sum_{j=1}^k \frac{1}{\sqrt{2\pi}\sigma_j} e^{-\frac{(o_i - \mu_j)^2}{2\sigma_j^2}}$$